

Taehun Ryu

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Experience

Graduate Research Intern

Jul 2024 – Feb 2025

3D Vision & Robotics Lab, Advisor: Prof. Kyungdon Joo

Education

Ulsan National Institute of Science and Technology (UNIST)

Mar 2025 – Present

M.S-Ph.D. integrated course in Artificial Intelligence

GPA: 4.05 / 4.3

Hanyang University ERICA

Mar 2019 – Aug 2024

B.S. in Robotics

GPA: 4.18 / 4.5

Publications

- [1] Taehun Ryu, Changwoo Kang, and Kyungdon Joo. **From Corners to Fiducial Tags: Revisiting Checkerboard Calibration for Event Cameras**. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2026.

Projects

Stereo Visual SLAM Enhancement

Jul. 2024 – Nov. 2024

3D Vision & Robotics Lab, UNIST

- Integrated SSC-based adaptive non-maximal suppression into ORB-SLAM3 to improve spatial feature coverage and tracking stability..

Table Robot System — Multi-Robot Localization and Perception

Sep. 2023 – Jun. 2024

Hanyang University ERICA

- Developed an onboard localization pipeline fusing LiDAR odometry, wheel encoders, and IMU measurements with an extended Kalman filter.
- Built a ceiling-camera-based robot tracking system and fused external observations with onboard localization using confidence-weighted position estimates.

Large Unmanned Store System — Multi-Camera Robot Perception

Sep. 2022 – Jan. 2023

Hanyang University ERICA

- Calibrated multiple ceiling-mounted cameras and transformed their observations into a shared global coordinate frame for customer localization.
- Developed a customer-following robot perception system using multi-camera tracking, LiDAR-based obstacle detection, and ArUco landmarks for odometry drift correction.

Awards

- **Gold Prize** — College of Engineering Capstone Design Competition Dec 2024
- **Third Prize** — National Intelligent Robotics Capstone Design Challenge Nov 2024
- **Undergraduate Student Paper Award** — Institute of Control, Robotics and Systems, for the paper “Table Robot System for Complete Restaurant Hall Automation” Jul 2024
- **Grand Prize** — Robotics Engineering Capstone Design Competition Jun 2024
- **Third Prize** — Digital Innovation Shared University Robotics Competition Jan 2023

Patents

- System and Method for Controlling a Table Robot (Korean Patent Application No. 10-2025-0008102, Patent Pending) Jan 2025

Scholarships

- National Excellence Scholarship in Science and Engineering Fall 2023
- Hanyang ERICA Academic Excellence Scholarship Spring 2024
- SHARE Academic Excellence Scholarship I Spring 2024, Fall 2022
- Special Academic Excellence Scholarship Fall 2023, Fall 2022

Skills

- **Programming:** C/C++, Python
- **Robotics:** ROS2
- **Tools:** Git, Docker
- **Certification:** Computer Specialist in Spreadsheet & Database Level-1